

Practical 5 & 6

General purpose utilities in linux PATH, man, echo, printf, script, passwd, uname, who, date, stty, pwd, man, echo, bc

Working with Directories and files : A.ls, mkdir, rmdir, B. file, touch, rm, cp. mv, file, touch, rm, cp. mv, rename

1] CREATING A FILE:

```
student@student-virtual-machine:~$ cat> file1
operating system
computer system
user
^C
```

2] ADDING MORE CONTENT TO FILE :

```
student@student-virtual-machine:~$ cat file1
examples;
-windows
-linux
-android
-macOS
-ios
examples:
-linux
-windows
-android
-macOS
-ios
```

3] COMMAND “WHO” :

```
student@student-virtual-machine:~$ who
student  tty2          2026-07-14 19:03 (tty2)
```

4] COMMAND “PWD” :

```
student@student-virtual-machine:~$ pwd
/home/student
```

5]COMMAND ‘LS’:

```
student@student-virtual-machine:~$ ls
Desktop  Documents  Downloads  file1  Music  Pictures  Public  snap  Templates  Videos
```

6] MAKING OF DIRECTORY:

```
student@student-virtual-machine:~$ mkdir d1
```

7]INODE NUMBERS:

```
student@student-virtual-machine:~$ ls -l
655986 d1          655878 Documents  655962 file1   655880 Pictures  655789 snap      655881 Videos
655874 Desktop    655875 Downloads  655879 Music   655877 Public    655876 Templates
```

8]DATE:

```
student@student-virtual-machine:~$ date
Tuesday 14 July 2026 07:28:39 PM IST
```

9] PS:

```
student@student-virtual-machine:~$ ps
  PID TTY          TIME CMD
 2689 pts/0        00:00:00 bash
 2975 pts/0        00:00:00 ps
```

10] PS -L

```
student@student-virtual-machine:~$ ps -l
 F S   UID        PID      PPID  C PRI  NI ADDR SZ WCHAN  TTY          TIME CMD
 0 S   1000        2689      2671  0  80   0 -  2783 do_wai pts/0        00:00:00 bash
 0 R   1000        2976      2689  0  80   0 -  3167 -          pts/0        00:00:00 ps
```

11] bc (BASH CALCULATOR)

```
student@student-virtual-machine:~$ bc
bc 1.07.1
Copyright 1991-1994, 1997, 1998, 2000, 2004, 2006, 2008, 2012-2017 Free Software Foundation, Inc.
This is free software with ABSOLUTELY NO WARRANTY.
For details type 'warranty'.
4+5
9
5-5
0
```

CONVERSION OF BINARY TO NUMBER

```
ibase=2
17/2
1
```

CONVERSION DECIMAL TO BINARY

```
obase=2
17/5
0
```

12] MAN STTY

```

TTY(1)                                     User Commands
NAME
  stty - change and print terminal line settings

SYNOPSIS
  stty [-F DEVICE | --file=DEVICE] [SETTING]...
  stty [-F DEVICE | --file=DEVICE] [-a|--all]
  stty [-F DEVICE | --file=DEVICE] [-g|--save]

DESCRIPTION
  Print or change terminal characteristics.

  Mandatory arguments to long options are mandatory for short options too.

  -a, --all
      print all current settings in human-readable form

  -g, --save
      print all current settings in a stty-readable form

  -F, --file=DEVICE
      open and use the specified DEVICE instead of stdin

  --help display this help and exit

  --version
      output version information and exit

```

Manual page stty(1) line 1 (press h for help or q to quit)

13] STTY -a

```

student@student-virtual-machine:~$ stty -a
speed 38400 baud; rows 30; columns 120; line = 0;
intr = ^C; quit = ^\; erase = ^H; kill = ^U; eof = ^D; eol = <undef>; eol2 = <undef>; swtch = <undef>; start = ^Q;
stop = ^S; susp = ^Z; rprnt = ^R; werase = ^W; lnext = ^V; discard = ^O; min = 1; time = 0;
-parenb -parodd -cmspar cs8 -hupcl -cstopb cread -cllocal -crtsets
-ignbrk -brkint -ignpar -parmrk -inpck -istrip -inlcr -igncr icrnl ixon -ixoff -iuc1c -ixany -lnxbel iutf8
bpost -olcuc -ocrnl onlcr -onocr -onlret -ofill -ofdel nl0 cr0 tab0 bs0 vt0 ff0
tsiq icanon isexten echo echoe echok -echonl -noflsh -xcase -tostop -echoprnt echoctl echoke -flusho -extproc
student@student-virtual-machine:~$

```

14] PASSWD

```

student@student-virtual-machine:~$ passwd
Changing password for student.
Current password:
New password:
BAD PASSWORD: The password is shorter than 8 characters
New password:
Password change has been aborted.
passwd: Authentication token manipulation error
passwd: password unchanged

```

15] UNAME

```

student@student-virtual-machine:~$ uname
Linux

```

16] UNAME -s

```
student@student-virtual-machine:~$ uname -s
Linux
```

17]PRINTF “ANUSHREE”

```
student@student-virtual-machine:~$ printf "anushree"
anushreestudent@student-virtual-machine:~$
```

18] LS -L (LIST ALL CHARARCTERS AND ALSO EXPLAIN)

```
student@student-virtual-machine:~$ ls -l
total 44
drwxrwxr-x 2 student student 4096 Jul 14 19:24 d1
drwxr-xr-x 3 student student 4096 Jul 10 21:57 Desktop
drwxr-xr-x 2 student student 4096 Jul 10 20:46 Documents
drwxr-xr-x 2 student student 4096 Jul 10 20:46 Downloads
-rw-rw-r-- 1 student student  94 Jul 14 19:20 file1
drwxr-xr-x 2 student student 4096 Jul 10 20:46 Music
drwxr-xr-x 3 student student 4096 Jul 14 19:15 Pictures
drwxr-xr-x 2 student student 4096 Jul 10 20:46 Public
drwx----- 3 student student 4096 Jul 10 20:46 snap
drwxr-xr-x 2 student student 4096 Jul 10 20:46 Templates
drwxr-xr-x 2 student student 4096 Jul 10 20:46 Videos
student@student-virtual-machine:~$
```

19] LS – a

```
student@student-virtual-machine:~$ ls -a
.  .bash_history  .bashrc  .config  Desktop  Downloads  .lessht  Music  .profile  snap  Videos
.. .bash_logout  .cache  d1  Documents  file1  .local  Pictures  Public  Templates
student@student-virtual-machine:~$
```

20] LS -D

```
student@student-virtual-machine:~$ ls -d
.
```

21] LS-A

```
student@student-virtual-machine:~$ ls -A
.bash_history  .bashrc  .config  Desktop  Downloads  .lessht  Music  .profile  snap  Videos
.bash_logout  .cache  d1  Documents  file1  .local  Pictures  Public  Templates
student@student-virtual-machine:~$
```

22] LS -r

```
student@student-virtual-machine:~$ ls -r
Videos Templates snap Public Pictures Music file1 Downloads Documents Desktop d1
student@student-virtual-machine:~$
```

23] LS -R

```
student@student-virtual-machine:~$ ls -R
.:
d1 Desktop Documents Downloads file1 Music Pictures Public snap Templates Videos

./d1:

./Desktop:
class syainl test typescript

./Desktop/syainl:
ainl huda

./Documents:

./Downloads:

./Music:

./Pictures:
Screenshots

./Pictures/Screenshots:
'Screenshot from 2026-07-14 19-15-15.png' 'Screenshot from 2026-07-14 19-16-31.png'

./Public:

./snap:
snapd-desktop-integration

./snap/snapd-desktop-integration:
```

24]LS -s

```
student@student-virtual-machine:~$ ls -s
total 44
4 d1 4 Desktop 4 Documents 4 Downloads 4 file1 4 Music 4 Pictures 4 Public 4 snap 4 Templates 4 Videos
student@student-virtual-machine:~$
```

25] LS -S

```
student@student-virtual-machine:~$ ls -S
d1 Desktop Documents Downloads Music Pictures Public snap Templates Videos file1
student@student-virtual-machine:~$
```

26] LS -t

```
student@student-virtual-machine:~$ ls -t
d1 file1 Pictures Desktop Documents Downloads Music Public Templates Videos snap
student@student-virtual-machine:~$
```

27] LS t*

```
student@student-virtual-machine:~$ ls t*
test1 test2 test3
student@student-virtual-machine:~$
```

28] ls te*

```
student@student-virtual-machine:~$ ls te*
test1 test2 test3
student@student-virtual-machine:~$
```

29] wc test1

```
student@student-virtual-machine:~$ w test1
 20:23:13 up  1:20,  1 user,  load average: 0.00, 0.02, 0.28
USER      TTY      FROM              LOGIN@   IDLE   JCPU   PCPU   WHAT
student@student-virtual-machine:~$ wc test2
 3  3 19 test2
student@student-virtual-machine:~$ wc test3
 3  3  9 test3
student@student-virtual-machine:~$ wc test1
 1  1 10 test1
```

30]who -Hu

```
student@student-virtual-machine:~$ who -Hu
NAME      LINE      TIME                IDLE          PID COMMENT
student   tty2       2026-07-14 19:03 00:34         1607 (tty2)
```

31]whoami

```
student@student-virtual-machine:~$ who am i
student@student-virtual-machine:~$ whoami
student
```

32] echo \$HOME

```
student@student-virtual-machine:~$ echo $HOME
/home/student
```

33]

```
student@student-virtual-machine:~$ cd syaiml1/syaiml2
student@student-virtual-machine:~/syaiml1/syaiml2$
```

34]

```
student@student-virtual-machine:~$ mkdir syaiml1/syaiml2/syaiml3
student@student-virtual-machine:~$
```

35]

```
student@student-virtual-machine:~$ rmdir syaiml1
rmdir: failed to remove 'syaiml1': Directory not empty
```

36. Touch

```
student@student-virtual-machine:~$ touch file1 file2 file3
student@student-virtual-machine:~$ ls
Desktop Documents Downloads file1 file2 file3 Music Pictures Public snap Templates Videos
```

37. File

```
student@student-virtual-machine:~$ file typescript
typescript: ASCII text, with CRLF, CR, LF line terminators, with escape sequences
```

38. Copy (cp)

```
student@student-virtual-machine:~$ cp file1 file1copy
student@student-virtual-machine:~$ ls
Desktop Documents Downloads file1 file1copy file2 file3 Music Pictures Public snap Templates typescript Videos
```

Copy in interactive way (asking permission for copy) :

```
student@student-virtual-machine:~$ cp -i file1 file1copy
cp: overwrite 'file1copy'? yes
```

Multiple files copy

```
student@student-virtual-machine:~$ mkdir test1
```

```
student@student-virtual-machine:~$ cp file1 file1copy test1
```

```
student@student-virtual-machine:~$ cd test1
student@student-virtual-machine:~/test1$ ls
file1 file1copy
```

Entire directory copy

```
student@student-virtual-machine:~$ mkdir test1/test2
```

```
student@student-virtual-machine:~$ cat > test1/test2/file3
hello this is file 3, using absolute path
this is cool
^C
```

```
student@student-virtual-machine:~$ cp -R test1 dircopy
```

39. Nv

```
student@student-virtual-machine:~$ nv file1 file22
```

40. Rm

```
student@student-virtual-machine:~$ rm -i file3
rm: remove regular empty file 'file3'? yes
```

Rm on multiple files

```
student@student-virtual-machine:~$ rm -i f1 f2 f3
rm: remove regular empty file 'f1'? yes
rm: remove regular empty file 'f2'? yes
rm: remove regular empty file 'f3'? _yes
```

```
student@student-virtual-machine:~$ touch t1 t2 t3
student@student-virtual-machine:~$ rm -r t*
```

41. Rmdir

If dir is empty

```
student@student-virtual-machine:~$ mkdir test4
student@student-virtual-machine:~$ rmdir test4
```

42. PATH

```
ubuntu@ubuntu:~/SYAIML$ echo $PATH
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin:/snap/bin
```

43. man

```
ubuntu@ubuntu:~/SYAIML -- man ls
LS(1)                                     General Commands Manual                                     LS(1)

NAME
    ls - List directory contents.  Ignore files and directories starting with a '.' by default

SYNOPSIS
    ls [--help] [--format] [-C] [-l|--long] [-x] [-T|--tabsize] [-n] [--zero] [-D|--dired] [--hyperlink] [-i]
    [-o] [-g] [-n|--numeric-uid-gid] [--quoting-style] [-N|--literal] [-b|--escape] [-Q|--quote-name]
    [-q|--hide-control-chars] [--show-control-chars] [--time] [-c] [-u] [--hide] [-I|--ignore] [-B|--ignore-back-
    ups] [--sort] [-S] [-t] [-v] [-X] [-U] [-L|--dereference] [--dereference-command-line-synlink-to-dir]
    [-H|--dereference-command-line] [-G|--no-group] [--author] [-a|--all] [-A|--almost-all] [-f] [-d|--directory]
    [-h|--human-readable] [-k|--kibibytes] [--si] [--block-size] [-i|--inode] [-r|--reverse] [-R|--recursive]
    [-w|--width] [-s|--size] [--color] [--indicator-style] [-F|--classify] [--file-type] [-p] [--tine-style]
    [--full-time] [-Z|--context] [--group-directories-first] [-V|--version] [paths]

DESCRIPTION
    List directory contents.  Ignore files and directories starting with a '.' by default

OPTIONS
    Help  --help Print help information.

    --format
        Set the display format.

    -C    Display the files in columns.

    -l, --long
        Display detailed information.

    -x    List entries in rows instead of in columns.

Manual page ls(1) line 1 (press h for help or q to quit)
```

44. echo

```
ubuntu@ubuntu:~/SYAIML$ echo $HOME
/home/ubuntu
```

```
ubuntu@ubuntu:~/SYAIML$ echo "Hello Linux Hows life"
Hello Linux Hows life
```

45. printf

```
ubuntu@ubuntu:~/SYAIML$ printf "Welcome to Linux\n This is how print works\n"
Welcome to Linux
This is how print works_
```

46. script

```
ubuntu@ubuntu:~/SYAIML$ script mysession.txt
Script started, output log file is 'mysession.txt'.
ubuntu@ubuntu:~/SYAIML$ pwd
/home/ubuntu/SYAIML
ubuntu@ubuntu:~/SYAIML$ date
Tue Jul 14 11:55:30 AM UTC 2026
ubuntu@ubuntu:~/SYAIML$ echo "recording complete"
recording complete
ubuntu@ubuntu:~/SYAIML$ exit
exit
Script done.
```

47. passwd

```
ubuntu@ubuntu:~/SYAIML$ passwd --help
Usage: passwd [options] [LOGIN]

Options:
  -a, --all                report password status on all accounts
  -d, --delete             delete the password for the named account
  -e, --expire             force expire the password for the named account
  -h, --help              display this help message and exit
  -k, --keep-tokens        change password only if expired
  -i, --inactive INACTIVE set password inactive after expiration
                           to INACTIVE
  -l, --lock               lock the password of the named account
  -n, --mindays MIN_DAYS  set minimum number of days before password
                           change to MIN_DAYS
  -q, --quiet              quiet mode
  -r, --repository REPOSITORY change password in REPOSITORY repository
  -R, --root CHROOT_DIR   directory to chroot into
  -P, --prefix PREFIX_DIR directory prefix
  -S, --status             report password status on the named account
  -u, --unlock            unlock the password of the named account
  -w, --warndays WARN_DAYS set expiration warning days to WARN_DAYS
  -x, --maxdays MAX_DAYS set maximum number of days before password
                           change to MAX_DAYS
  -s, --stdin             read new token from stdin
```

48. uname

```
ubuntu@ubuntu:~/SYAIML$ uname -a
Linux ubuntu 7.0.0-14-generic #14-Ubuntu SMP PREEMPT_DYNAMIC Mon Apr 13 11:09:53 UTC 2026 x86_64 GNU/Linux
```

49. Who

```
ubuntu@ubuntu:~/SYAIML$ who -Hu
```

NAME	LINE	TIME	IDLE	PID	COMMENT
ubuntu@ubuntu:~/SYAIML\$ who					
ubuntu@ubuntu:~/SYAIML\$ whoami					
ubuntu					

50. date

```
ubuntu@ubuntu:~/SYAIML$ date
Tue Jul 14 11:58:57 AM UTC 2026
```

51. stty

```
ubuntu@ubuntu:~/SYAIML$ stty -a
speed 38400 baud; rows 32; columns 120; line = 0;
intr = ^C; quit = ^\; erase = ^?; kill = ^U; eof = ^D; eol = <undef>; eol2 = <undef>; swtch = <undef>; start = ^Q;
stop = ^S; susp = ^Z; rprnt = ^R; werase = ^W; lnext = ^V; discard = ^O; min = 1; time = 0;
-parenb -parodd -cmspar cs8 -hupcl -cstopb cread -clocal -crtscts
-ignbrk -brkint -ignpar -parmrk -inpck -istrip -inlcr -igncr icrnl -ixoff -tandem ixon -ixany -imaxbel iutf8
opost -olcuc -ocrnl onlcr -onocr -onlret -ofdel nl0 cr0 tab0 bs0 vt0 ff0
isig icanon isexten echo echoe echok -echonl -noflsh -tostop -echoprnt echoctl echoke -flusho -extproc
```

52. pwd

```
ubuntu@ubuntu:~/SYAIML$ pwd
/home/ubuntu/SYAIML
```

53. Bc

```
ubuntu@ubuntu:~/SYAIML$ bc
bc 1.07.1
Copyright 1991-1994, 1997, 1998, 2000, 2004, 2006, 2008, 2012-2017 Free Software Foundation, Inc
This is free software with ABSOLUTELY NO WARRANTY.
For details type 'warranty'.
10+3
13
50/3
16
scale = 2
50/3
16.66
quit
```